25901008 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) REGULAR EXAMINATION NOVEMBER 2025

FIRST SEMESTER

SKILL DEVELOPMENT COMPONENTS (SDC) - MG1SDCMFT101 - PRINCIPLES OF HEALTH & FITNESS

(2025 ADMISSION ONWARDS)

Duration: 2 Hour **Maximum Marks:70**

Part A

Multiple Choice Questions

Answer any **10** questions

Each question carries **1** marks

1. Which of the following best defines muscular endurance? [U] /[CO1]

c) The ability of muscles to perform repeated contractions over time without fatigue

2. When designing a personalized fitness plan, how can the FITT principle guide the process effectively? [U] /[CO1]

b) It provides a structured approach to modifying intensity and type of activity

2. Which of the following is a safe practice when using supplements for athletes or active individuals? [A] /[CO3

b). Consulting a healthcare or sports nutrition professional before use

3. What is the main goal of stress management? [U] /[CO1]

b) To recognize stressors and develop effective coping strategies for maintaining balance and wellbeing

4. Which of the following is a major benefit of regular physical activity on physical health?

[U]/CO1]

b) It strengthens bones and muscles

6. What is the recommended strategy for athletes to maintain hydration during prolonged physical activity? [A] /[CO3]

b) Consume water before, during, and after exercise

7. Which of the following is a negative effect of inadequate sleep on fitness and health? [An]/[CO4]

c) Decreased reaction time and increased injury risk

8. When designing a fitness and wellness program for individuals recovering from substance abuse, which approach is most appropriate? [C] /[CO5]

a. Incorporate gradual physical activity, counseling, and balanced nutrition

9. Which of the following best defines the concept of health as per the World Health Organization (WHO)? [U] /[CO1

a) Incorporate gradual physical activity, counseling, and balanced nutrition

10. When designing a basic fitness program, how should assessment results be used? [U] [CO1]

a) To set realistic goals and track progress over time

11. To maintain proper hydration and support physical performance, what is the most effective strategy? [A] /[CO3]

b) Consume water consistently before, during, and after physical activity.

12. Lack of adequate sleep can lead to which of the following negative health outcomes? [An]/[CO4]

c) Decreased immune function and energy levels

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **2** marks

13. How do lifestyle choices influence overall health and well-being? Provide examples. [A] / [CO3]

Lifestyle choices directly affect overall health and well-being by influencing physical fitness, mental health, and disease risk. For example, regular exercise improves heart health, while healthy eating lowers the risk of obesity and diabetes. Choosing not to smoke also reduces the risk of lung and heart diseases.

14. Define dietary supplements and explain their general purpose in supporting health and fitness. [U] / [CO1]

Dietary supplements are products taken to add nutrients to the diet, such as vitamins, minerals, or proteins. Their general purpose is to fill nutritional gaps, support overall health, and enhance fitness by improving energy, recovery, and performance when needed.

15. Why is it important to link goal setting with fitness assessment results when designing an exercise program?[U] /[CO1]

Linking goal setting with fitness assessment results is important because it helps create realistic, personalized, and achievable exercise plans. Fitness assessments show your current level, and goals based on these results guide proper training and track progress effectively.

16. Define physical inactivity and explain how it differs from a sedentary lifestyle. [U] [CO1]

Physical inactivity means not meeting the recommended levels of physical activity, such as not exercising enough. A sedentary lifestyle involves sitting or lying down for long periods (e.g., screen time), even if a person does some exercise.

17. Define a sedentary lifestyle and explain its impact on overall health. [U] / [CO1]

A sedentary lifestyle is a way of living that involves long periods of sitting or minimal movement. It negatively impacts overall health by increasing the risk of obesity, heart disease, diabetes, and reduced physical fitness.

18. Explain the difference between physical fitness and overall health. [U] / [CO1]

Physical fitness refers to the body’s ability to perform physical activities efficiently, such as

strength, endurance, and flexibility. Overall health includes physical, mental, and social well-being, not just the ability to exercise.

19. List two key functions of vitamins and minerals in maintaining overall health. [K] / [CO1]

Vitamins and minerals support overall health by boosting the immune system and helping the body produce energy and maintain strong bones.

Part C

Short Essay

Answer any 4 questions Each question carries 5 marks

20.Discuss each of the five dimensions of health (physical, mental, emotional, social, and / spiritual) and explain their role in achieving overall well-being. [S][CO1]

Health is a multidimensional concept that includes physical, mental, emotional, social, and spiritual aspects. Physical health refers to the proper functioning of the body and includes fitness, nutrition, sleep, and freedom from disease. When physical health is maintained, the body has enough strength, stamina, and immunity to perform daily activities efficiently.

Mental health involves the ability to think clearly, make decisions, learn, and concentrate. A mentally healthy person can handle challenges, solve problems, and adapt positively to different situations.

Emotional health is the ability to understand, express, and manage one's feelings. It helps individuals cope with stress, maintain self-control, and build resilience during difficult moments.

Social health refers to building and maintaining healthy relationships, communicating effectively, and having a support system. Strong social connections reduce loneliness, increase confidence, and contribute to overall happiness.

Spiritual health involves inner peace, purpose, values, and belief systems. It gives meaning to life and promotes harmony, compassion, and positive attitudes.

These five dimensions are interconnected. When all are balanced, an individual experiences complete well-being and leads a healthy, fulfilling

21.Explain the concept of a balanced diet and its significance in maintaining health and physical fitness. How does it contribute to energy, growth, and overall well-being?[U] /[CO2]

A balanced diet is one that contains all essential nutrients—carbohydrates, proteins, fats, vitamins, minerals, fiber, and water—in the right quantities. It ensures the body receives everything needed for

proper growth, development, and daily functioning.

The significance of a balanced diet lies in its role in maintaining good health and physical fitness. Carbohydrates and fats provide energy required for movement, exercise, and daily activities. Proteins support muscle development, tissue repair, and overall growth, especially in children and athletes. Vitamins and minerals help regulate body processes, maintain immunity, support bone health, and prevent nutritional deficiencies. Fiber improves digestion, while water maintains hydration and helps regulate body temperature.

A balanced diet supports overall well-being by improving energy levels, strengthening immunity, preventing lifestyle diseases like obesity and diabetes, and keeping the body fit and active. It ensures better physical performance, faster recovery, and long-term health.

22.Explain how substance abuse and addictions can impact overall health and physical fitness. How do these behaviors affect energy, endurance, and recovery? U] /[CO2]

Substance abuse refers to the harmful use of alcohol, tobacco, drugs, or other addictive substances. These behaviors have serious negative effects on overall health and significantly reduce a person's physical fitness. When substances are used regularly, they damage vital organs such as the heart, lungs, liver, and brain. This leads to weakened immunity, digestive problems, hormonal imbalance, and an increased risk of chronic diseases such as cancer, heart disease, and respiratory disorders. Addiction also affects mental health by causing anxiety, depression, irritability, and poor concentration.

In terms of physical fitness, substance abuse greatly reduces the body's natural ability to perform. Energy levels drop because substances interfere with nutrient absorption, disturb sleep, and disrupt normal metabolic processes. Alcohol and drugs weaken the body's ability to convert food into usable energy, making individuals feel tired, weak, and less active.

Substance abuse also affects endurance, which is essential for physical activities and exercise. Smoking decreases lung capacity, reduces oxygen supply, and increases breathlessness. Alcohol and drugs slow down the cardiovascular system, making it difficult for the heart to pump blood efficiently. As a result, stamina decreases, exercise becomes difficult, and performance levels decline sharply.

Another major impact of addictions is on recovery. The body depends on proper rest, hydration, and nutrition to repair muscles and tissues after physical activity. However, substances slow down healing, increase inflammation, and disturb sleep patterns. Alcohol dehydrates the body, while drugs disturb hormone production, both of which delay recovery. This increases the risk of injuries and makes the body more vulnerable to fatigue.

23..Analyze the long-term health risks associated with a sedentary lifestyle. Include both physical and psychological consequence [An] [CO4]

A sedentary lifestyle, characterized by prolonged sitting and minimal physical activity, poses serious long-term health risks that affect both physical and psychological well-being. Physically, inactivity slows down the body's metabolic processes, making it easier to accumulate excess body fat and increasing the risk of obesity. Over time, this leads to a higher likelihood of developing chronic conditions such as type 2 diabetes, hypertension, and cardiovascular diseases. Lack of regular movement weakens the muscles and reduces bone density, contributing to osteoporosis and greater susceptibility to fractures. Joint stiffness, poor posture, and chronic back or neck pain become common as the body loses flexibility and strength. Sedentary habits also impair blood circulation, raising the risk of blood clots and varicose veins. Furthermore, inactivity may weaken the immune system, making the body less capable of fighting infections.

Psychologically, a sedentary lifestyle can negatively affect mental health. Limited physical activity reduces the release of endorphins, dopamine, and serotonin—hormones responsible for improving mood and reducing stress. As a result, individuals may experience increased levels of anxiety, irritability, and depression. Excessive screen time and isolation further contribute to feelings of loneliness and reduced social engagement. Over time, these psychological effects can diminish self-esteem and motivation, creating a cycle where low mood leads to even less activity. Cognitive decline is another potential risk, as regular movement is known to improve memory, attention, and brain function.

In the long run, the combined physical and psychological consequences of a sedentary lifestyle can significantly lower one's quality of life. They increase the risk of disability, reduce life expectancy, and limit daily functioning. Therefore, incorporating regular exercise, reducing sitting time, and maintaining an active routine are essential steps to prevent these long-term health risks and promote overall well-being

24..Identify and discuss at least three common causes of stress in daily life. Explain how each can influence both mental and physical health. (An) CO4

Stress is a natural response to challenges or demands in daily life, but when it becomes frequent or intense, it can affect both mental and physical health. Several common factors contribute to stress, and three of the most significant are academic/work pressure, financial problems, and relationship conflicts Academic or work pressure is one of the most widespread sources of stress. Students face deadlines, exams, and performance expectations, while employees deal with heavy workload, long hours, and job responsibilities. Mentally, this pressure causes anxiety, reduced concentration, irritability, and lack of motivation. Physically, it can lead to headaches, fatigue, increased heart rate, muscle tension, and disturbed sleep. Over time, it may contribute to burnout Financial difficulties—such as debts, rising expenses, or job insecurity—create constant worry and fear about

the future. Mentally, financial stress leads to overthinking, restlessness, and feelings of helplessness. Physically, it triggers high blood pressure, digestive problems, weakened immunity, and irregular sleep patterns. Prolonged financial stress can also contribute to chronic illnesses due to continuous strain on the body.

Relationship problems with family, friends, or colleagues also create emotional stress. Conflicts, misunderstandings, or lack of support can disturb mental peace. Mentally, relationship stress causes mood swings, sadness, anger, and emotional exhaustion. Physically, it can result in stomach issues, loss of appetite, rapid heartbeat, and low energy levels. Social stress also increases loneliness, which further affects mental well-being.

25..Explain the role of safe food handling and hygiene practices in preventing illness and promoting overall health. How does maintaining food safety contribute to sustaining an active lifestyle? [A] / [CO3]

Safe food handling and hygiene practices play a crucial role in protecting individuals from food-borne diseases and promoting overall health. These practices include proper washing of hands, cleaning utensils, cooking food at the right temperature, storing food safely, and preventing contamination from raw ingredients. When food is handled hygienically, harmful microorganisms such as bacteria, viruses, and parasites are eliminated or prevented from multiplying. This reduces the risk of illnesses like food poisoning, diarrhea, stomach infections, and other digestive problems.

Safe food handling also helps preserve the nutritional value of food. Proper washing and cooking techniques ensure that nutrients are maintained, allowing the body to receive essential vitamins and minerals needed for daily functioning. Hygiene practices like timely refrigeration prevent spoilage and protect the body from toxins that can harm vital organs. As a result, the immune system stays strong, digestion improves, and the overall health of an individual remains stable.

Maintaining food safety directly supports an active lifestyle. When individuals are healthy and free from frequent infections, they have more energy, strength, and stamina to participate in physical activities, exercise, work, and daily responsibilities. Good nutrition from safe food promotes faster recovery, better muscle function, and sustained energy levels. On the other hand, consuming contaminated or poorly handled food can lead to sickness, weakness, dehydration, and fatigue, all of which interrupt physical activity and reduce performance.

Part D

Answer any 2 questions

Each question carries 15 marks

26.Explain the purpose and importance of using fitness assessment tools in evaluating health-related and skill-related fitness. How do these assessments help individuals understand their current fitness levels and guide improvements? [U] / [CO1]

Fitness assessment tools serve as essential instruments in evaluating both health-related and skill-related fitness by providing objective, measurable, and scientific information about an individual's physical capabilities, strengths, and limitations, thereby forming the foundation for effective fitness planning, performance enhancement, and long-term health maintenance. These assessments encompass a wide range of tests, including measurements of cardiovascular endurance, muscular strength, muscular endurance, flexibility, and body composition for health-related fitness, as well as agility, balance, coordination, power, speed, and reaction time for skill-related fitness, all of which together offer a clear and holistic understanding of one's overall fitness status. The primary purpose of using fitness assessment tools is to establish a baseline or starting point that reflects the individual's current physical condition, allowing for a detailed analysis of which areas are strong and which require improvement. Without such baseline data, individuals and fitness professionals would be unable to design safe and effective exercise programs tailored to personal needs, as assumptions about fitness levels often lead to inappropriate training intensity, increased injury risk, or ineffective training outcomes. By clearly identifying weaknesses—for instance, low muscular endurance, limited flexibility, or poor balance—fitness assessments guide individuals toward selecting appropriate exercises and training methods that specifically target those limitations, ensuring that efforts are directed where they are most needed. In addition, these assessment tools play a crucial role in goal-setting, helping individuals create realistic, specific, and measurable goals based on factual data rather than guesswork, thus promoting motivation and long-term commitment. When individuals see quantifiable results, such as improved endurance scores or increased strength measurements, they gain a sense of progress and achievement that reinforces their motivation to continue exercising. Moreover, regular assessments help monitor progress over weeks or months, making it possible to evaluate the effectiveness of the training program and make necessary adjustments to maintain continuous improvement. For instance, if body composition measurements reveal minimal changes despite consistent efforts, trainers can modify nutritional plans or training intensity to achieve better results. Fitness assessments are also vital in identifying potential health risks such as high body fat percentage, low aerobic endurance, or muscular imbalances, which could contribute to chronic diseases, poor posture, or susceptibility to injuries. Early identification of these risks empowers individuals to take preventive action, adopt healthier lifestyle habits, and engage in corrective exercises that promote long-term wellness. In sports and physical education, skill-related fitness assessments are particularly important because they help athletes understand the specific physical skills required for optimal performance in their chosen sport. For example, testing agility, power, and speed allows athletes to refine their training according to their sport's demands, while also helping coaches identify talent and place individuals in roles that suit their natural abilities. These assessments not only enhance athletic performance but also contribute to better movement

efficiency, improved coordination, and reduced risk of injury during high-intensity activities. Furthermore, fitness assessment tools promote self-awareness by providing individuals with a clear picture of their actual fitness levels, helping them understand that fitness involves multiple components and not just appearance or body weight.

This awareness fosters a more balanced and holistic approach to health, encouraging individuals to work on areas they might have previously ignored, such as flexibility training, cardiovascular endurance, or balance exercises. By giving people insight into their physical strengths and weaknesses, fitness assessments encourage education, responsibility, and active participation in their own fitness journey. In addition, these tools help trainers and educators design fitness programs that are safe for beginners, appropriate for special populations (such as older adults or individuals with medical conditions), and adaptable to different fitness levels. For example, a person with low functional balance may be guided toward stability exercises before attempting more dynamic movements, preventing falls or injuries. The importance of fitness assessment tools also lies in their ability to promote long-term behavioral change and healthy lifestyle development. When individuals receive continuous feedback about their progress, they become more mindful of their habits, make healthier choices, and gradually develop intrinsic motivation to maintain an active lifestyle. This sustained commitment is essential for preventing lifestyle diseases, improving mental well-being, and enhancing quality of life. Moreover, fitness assessments provide valuable data for scientific research, health monitoring, and community fitness programs, contributing to improved public health strategies and educational practices.

27. Explain the importance of proper nutrition for active individuals in maintaining energy levels, enhancing performance, and supporting recovery. Discuss how the balance of carbohydrates, proteins, and fats, along with hydration, influences physical performance and overall health.

[U] / [CO1, CO3, CO4]

Proper nutrition plays a central role in maintaining the health, performance, and overall well-being of active individuals, whether they are athletes, fitness enthusiasts, or people engaged in regular physical work. The body relies heavily on nutrients for fuel, tissue repair, metabolic functions, and recovery, making balanced nutrition essential for achieving optimal physical performance. For active individuals, the demands placed on the body are significantly higher than those of a sedentary person. This increased demand requires a thoughtful and strategic intake of macronutrients—carbohydrates, proteins, and fats—as well as micronutrients, vitamins, minerals, and hydration. Each of these components contributes uniquely to energy production, muscle strength, endurance, recovery, and overall physical health. Without adequate nutritional support, physical performance declines, fatigue sets in, recovery slows, and the risk of injury increases. Thus, proper nutrition is not an option but a requirement for anyone involved in regular physical activity.

Carbohydrates serve as the primary and most efficient source of energy for active individuals. They are broken down into glucose, which fuels muscle contractions and supports both aerobic and anaerobic activities. During moderate to high-intensity workouts, the body depends heavily on stored glycogen in muscles and the liver. When carbohydrate intake is insufficient, glycogen stores become depleted quickly, leading to early fatigue, reduced endurance, and decreased performance. This is especially problematic during long-duration activities such as running, cycling, or team sports. Consuming adequate carbohydrates before exercise provides readily available energy, while intake during prolonged activities helps maintain blood glucose levels and delay exhaustion. Post-workout carbohydrates are equally important because they replenish lost glycogen and speed up recovery. Complex carbohydrates—such as whole grains, oats, quinoa, brown rice, fruits, and vegetables—supply long-lasting energy, fiber, and essential nutrients that support overall health. Therefore, carbohydrates form the backbone of an active individual's diet, enabling consistent performance, stable energy levels, and efficient recovery.

Proteins are vital for muscle repair, growth, and recovery. During exercise, especially resistance and high-intensity training, muscle fibers undergo microscopic damage. To repair these fibers and stimulate muscle growth, the body requires adequate protein intake. Proteins supply essential amino acids that build and maintain muscle tissue, strengthen the immune system, produce hormones, and support enzymatic reactions.

Hydration is another crucial element in supporting physical performance and overall well-being. Water makes up more than half of the body's composition and is essential for temperature regulation, nutrient transport, joint lubrication, and waste removal. During physical activity, the body loses water and electrolytes through sweat. If these lost fluids are not replaced, dehydration occurs, leading to a decline in performance, fatigue, dizziness, muscle cramps, and in severe cases, heat exhaustion or heatstroke. Even mild dehydration can reduce endurance and impair concentration and reaction time.

The balance of carbohydrates, proteins, fats, and hydration significantly influences physical performance and well-being. Carbohydrates supply immediate energy for muscle contractions, proteins repair and strengthen muscles, and fats provide endurance and support vital body processes. Without the right balance, performance becomes inconsistent, recovery slows, and energy levels fluctuate. For instance, a diet low in carbohydrates can cause glycogen depletion, resulting in rapid fatigue and poor endurance.balanced and nutrient-dense diet to sustain peak physical condition, prevent injuries, and maintain overall wellness.

Beyond physical performance, proper nutrition affects mental clarity, mood, and long-term health outcomes. Nutrients such as omega-3 fatty acids, vitamins, and minerals support brain function, reduce stress, and promote emotional stability. Inadequate nutrition weakens immunity, increases the risk of illness, and disrupts metabolic processes, all of which interfere with training consistency and

performance. Active individuals who fuel their bodies correctly experience better concentration, faster reaction times, and improved motivation. Good nutrition also supports healthy body composition by regulating metabolism and maintaining muscle-to-fat ratios. Over time, these benefits contribute to improved athletic longevity, reduced injury risk, and enhanced overall quality of life.

28. Define what constitutes a healthy lifestyle and explain how it relates to the key principles of health and physical fitness. Why is adopting a healthy lifestyle essential for maintaining long-term wellness and preventing disease? [U] / [CO1]

A healthy lifestyle refers to a balanced and disciplined way of living that promotes physical, mental, emotional, and social well-being through positive daily habits. It includes conscious choices such as eating nutritious food, staying physically active, getting adequate sleep, managing stress, maintaining hygiene, nurturing relationships, avoiding harmful substances, and adopting regular health check-ups. A healthy lifestyle is not limited to the absence of illness; rather, it represents a proactive approach to enhancing the quality of life and functioning at one's best potential. It aligns closely with the key principles of health and physical fitness, which emphasize balance, consistency, prevention, and holistic well-being. In today's world, where lifestyle-related diseases are increasing rapidly due to sedentary habits, poor diet, and stress, adopting a healthy lifestyle has become essential for maintaining long-term health, preventing diseases, and ensuring overall wellness.

A healthy lifestyle is strongly connected to the principles of health, which include physical, mental, social, emotional, environmental, and spiritual well-being. Physically, it involves maintaining a strong and efficient body system through regular exercise, proper nutrition, and adequate rest. Mentally, it promotes clear thinking, emotional stability, stress management, and resilience. Socially, a healthy lifestyle supports meaningful interactions, positive relationships, and a supportive environment. Spiritually, it involves cultivating inner peace, purpose, and self-awareness. These principles work together to create balance in life, enabling individuals to function effectively and handle life's challenges. Without addressing any one of these aspects, overall health becomes incomplete. Thus, a healthy lifestyle becomes a practical application of these principles, ensuring harmony between the mind, body, and environment.

The principles of physical fitness—such as regular exercise, cardiovascular endurance, muscular strength, flexibility, and body composition—are also closely linked to a healthy lifestyle. Engaging in consistent physical activity improves the efficiency of the heart, lungs, and muscles while reducing the risk of obesity, diabetes, hypertension, and other chronic diseases. Strength training develops strong bones and muscles, enhancing functional movement and reducing the chances of injuries. Flexibility exercises maintain joint health, posture, and mobility, supporting daily tasks with

ease. Good body composition, achieved through balanced diet and activity, ensures an optimal ratio of fat to lean mass, which is crucial for metabolic health. Therefore, by adopting a healthy lifestyle that incorporates regular exercise and physical fitness practices, individuals enhance their physical capabilities, increase energy levels, and promote longevity.

One of the most important reasons a healthy lifestyle is essential is its role in disease prevention. Most modern health issues—such as obesity, cardiovascular diseases, type 2 diabetes, stroke, certain cancers, and mental health disorders—are directly linked to unhealthy habits. Poor diet, lack of physical activity, smoking, alcohol abuse, stress, irregular sleep, and pollution exposure contribute to long-term damage to bodily systems. By adopting healthy habits—like eating nutrient-dense foods, exercising regularly, avoiding tobacco and alcohol, practicing stress-reduction techniques, and maintaining proper sleep schedules—individuals can significantly reduce their risk of chronic diseases and enhance their body's natural defense systems. Preventive care is far more effective and less costly than medical treatment, which is why a healthy lifestyle forms the foundation of public health strategies worldwide.